BEU CONNECT

Bihar Engineering University, Patna

B.Tech 7th Semester Exam., 2021

Code: 102705 ***Time: 03 Hours***

Semester: VII ***Subject: AUTOMOBILE ENGINEERING*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven) : ***[2 x 7 = 14]***

(a) Ford EcoSport 2021 model has
- (i) manual transmission
- (ii) semi-automatic transmission
- (iii) fully-automatic transmission
- (iv) Both (i) and (iii)

(b) In a square type engine
- (i) geometrical shape is square
- (ii) diameter and length of piston are same
- (iii) two cylinders are placed horizontal and two vertical
- (iv) stroke length and cylinder bore are same

(c) The connecting rods are generally made of ____ shaped cross-section.
- (i) I
- (ii) C
- (iii) L
- (iv) H

(d) The vehicle ride will be comfortable if
- (i) unsprung mass is kept minimum
- (ii) sprung mass is kept minimum
- (iii) vehicle mass is kept minimum
- (iv) All of the above

(e) The tilting of the front wheels away from the vertical, when viewed from the front of car is called
- (i) toe in
- (ii) caster
- (iii) camber
- (iv) toe out

(f) The torque converter uses ____ to transfer torque.
- (i) air
- (ii) automatic transmission fluid (ATF)
- (iii) gears
- (iv) steel belt

(g) The starter motor is driven by
- (i) chain drive
- (ii) gear drive
- (iii) flat belt drive
- (iv) V-belt drive

(h) The aspect ratio (expressed in percentage) of the tyre is defined as the ratio of
- (i) section height to section width
- (ii) wheel diameter to section height
- (iii) wheel diameter to section width
- (iv) section width to section height

(i) In radial tyres
- (i) one-play layer runs diagonally one way and another layer runs diagonally the other way
- (ii) all plies run parallel to one another and vertical to tyre bead
- (iii) inner tubes are always used
- (iv) None of the above

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(j) The caster is called positive when the top of the pin is inclined to the

(i) rear of the vehicle
(ii) front of the vehicle
(iii) left of the vehicle
(iv) right of the vehicle

Q.2 ***[7+7=14]***

(a) With the help of neat sketch, mark different components of a 4-wheeler, 2 rear wheel-driven car.

(b) Distinguish between an articulated vehicle and a rigid vehicle. How do their construction and design differ from each other?

Q.3 ***[7+7=14]***

(a) What is common rail direct injection (CRDI) system? How is it different from conventional injection system?

(b) Differentiate between transistor-based coil ignition system and capacitive discharge ignition system.

Q.4 ***[7+7=14]***

(a) Classify different types of clutches. Describe the working of multiplate friction clutch.

(b) A cone clutch with a cone semi angle of 12° is to transmit 11.19 kW at 750 r.p.m. The width of the face is 1/4th of the mean diameter and the normal pressure between the contact faces is not to exceed 0.67 kgf/cm^2. Allowing for the coefficient of friction of 0.2, determine the main dimensions of the clutch and the axial force required.

Q.5 ***[8+6=14]***

(a) What is gearbox? Why do we need a gearbox in an automobile? State various functions of a gearbox.

(b) Sketch the layout of a synchromesh gearbox used on any Indian automobile.

Q.6 ***[8+6=14]***

(a) What is a torque converter? Sketch and explain the construction and working of a torque converter. In what respect is it different from a fluid flywheel?

(b) Explain the principle of working of an overdrive with suitable diagram.

Q.7 ***[7+7=14]***

(a) Sketch the assembly of a front wheel and a stub axle showing inclined king-pin and cambered wheel to provide center-point steering.

(b) With the help of a neat sketch, differentiate between king-pin inclination and steering-axis inclination.

Q.8 ***[8+6=14]***

(a) Differentiate between :
(i) Two leading shoe brake and two trailing shoe brake
(ii) Self-energizing and servo brakes

(b) What is tandem master cylinder? What are its advantages over an ordinary master cylinder?

Q.9 ***[8+6=14]***

(a) What is biodiesel? Explain various processes involved in transesterification process to convert vegetable oil into biodiesel.

(b) What are hybrid vehicles? With the help of a schematic diagram, explain the working of any hybrid electric auto-vehicle.

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